

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

CO5: Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – Data Frame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting

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2. Write a detailed pseudocode algorithm to compute statistical measures such as mean, median, standard deviation, and variance from a large dataset. The algorithm should demonstrate how to efficiently use array-based operations, apply filtering to exclude unwanted values, and perform aggregation. Include steps for validating the dataset and handling edge cases such as empty arrays or extreme outliers.

Input

A numerical dataset D.

Output

Mean, Median, Variance, and Standard Deviation.

Pseudocode

BEGIN

1. Read the dataset D.
2. Convert D into a numerical array.
3. Check whether the array is empty.
 IF array is empty THEN
 Display "Dataset is empty"
 STOP
END IF
4. Remove missing, non-numeric, and invalid values.
5. Filter unwanted values according to the given condition.
6. Check whether any valid values remain.
 IF no valid values remain THEN
 Display "No valid data available"

STOP

END IF

7. Calculate the mean: $\text{Mean} = \text{Sum of all valid values} / \text{Number of valid values}$

8. Sort the valid values.

9. Calculate the median:

IF number of values is odd THEN

Median = Middle value

ELSE

Median = Average of two middle values

END IF

10. Calculate the variance: $\text{Variance} = \text{Average of squared differences from the mean}$

11. Calculate standard deviation: $\text{Standard Deviation} = \text{Square Root of Variance}$

12. Identify extreme outliers using a suitable statistical method such as IQR or Z-score.

13. Exclude unwanted extreme outliers if required.

14. Recalculate the statistical measures after filtering.

15. Validate that all calculated results are finite and meaningful.

16. Display:

Mean

Median

Variance

Standard Deviation

END

Output

The algorithm produces the statistical measures of the validated and filtered dataset.

6. Design a pseudocode algorithm to perform data cleaning and transformation on time-series data. The algorithm should handle missing timestamps, interpolate missing values, remove outliers, and resample data at fixed intervals. Include steps for indexing by time, filtering invalid entries, and ensuring chronological consistency while maintaining efficiency.

Input

A time-series dataset containing timestamps and numerical values.

Output

A cleaned and regularly spaced time-series dataset.

Pseudocode

BEGIN

1. Read the time-series dataset D.
2. Convert the timestamp column into Date-Time format.
3. Identify and remove records having invalid timestamps.
4. Sort the dataset according to timestamp.
5. Set the timestamp column as the Data Frame index.
6. Check for missing timestamps.
7. Create a continuous time index at the required fixed time interval.
8. Reindex the dataset using the continuous time index.
9. Identify missing numerical values.
10. Interpolate missing values using time-based interpolation.
11. Calculate statistical limits for detecting outliers.
12. Identify values outside the acceptable limits.
13. Remove or replace the detected outliers.
14. Resample the time-series data at fixed intervals.
15. Apply an aggregation function such as mean to the resampled values.
16. Check that all timestamps are in chronological order.
17. Check that missing values have been handled.
18. Save the cleaned time-series dataset.
19. Display the final cleaned dataset.

END

Output

A chronologically ordered time-series dataset with missing values, invalid timestamps, and unwanted outliers handled.

9. Create a pseudocode algorithm to detect and handle outliers in a dataset using statistical methods such as Z-score or IQR. The algorithm should identify extreme values, decide whether to remove or cap them, and ensure that the resulting dataset maintains consistency and reliability for further analysis.

Input

A numerical dataset D.

Output

A dataset with identified and handled outliers.

Pseudocode

BEGIN

1. Read the dataset D.
2. Select the numerical columns.
3. Check for missing or invalid values.
4. Handle missing values before detecting outliers.
5. Select an outlier detection method: Z-SCORE or IQR.
6. IF Z-SCORE method is selected THEN Calculate the mean of the dataset.

 Calculate the standard deviation.

 FOR each value X in the dataset DO

 Calculate: $Z = (X - \text{Mean}) / \text{Standard Deviation}$

 IF absolute value of $Z > 3$ THEN

 Mark X as an outlier.

 END IF

 END FOR

ELSE IF IQR method is selected THEN

 Calculate first quartile Q1.

 Calculate third quartile Q3.

 Calculate: $IQR = Q3 - Q1$

 Calculate:

 Lower Limit = $Q1 - 1.5 \times IQR$

 Upper Limit = $Q3 + 1.5 \times IQR$

 FOR each value X DO

 IF $X < \text{Lower Limit}$ OR $X > \text{Upper Limit}$ THEN

 Mark X as an outlier.

 END IF

 END FOR

END IF

7. Display all detected outliers.
8. Choose an appropriate handling method:
 - Remove the outliers
 - OR
 - Cap the outliers at the boundary values.
9. Validate the resulting dataset.
10. Check that the dataset maintains consistence after outlier handling.
11. Save the cleaned dataset.
12. Display the final dataset.

END

Output

The algorithm identifies extreme values and produces a consistent dataset after removing or capping the outliers.

12. Write a pseudocode algorithm to implement dimensionality reduction using Principal Component Analysis (PCA). The algorithm should include steps for standardizing the dataset, computing covariance matrix, extracting eigenvalues and eigenvectors, and selecting principal components. Ensure efficient matrix operations and validation of reduced dimensions.

Input

A numerical dataset D and the required number of principal components K.

Output

A reduced-dimensional dataset containing K principal components.

Pseudocode

BEGIN

1. Read the dataset D.
2. Select the numerical features.
3. Handle missing or invalid values.
4. Check that the dataset contains sufficient numerical features for PCA.
5. Standardize each feature using:
 - $$\text{Standardized Value} = (\text{Value} - \text{Mean}) / \text{Standard Deviation}$$
6. Calculate the covariance matrix of the standardized dataset.
7. Calculate the eigenvalues and eigenvectors of the covariance matrix.

8. Sort the eigenvalues in descending order.
9. Arrange the corresponding eigenvectors according to the sorted eigenvalues.
10. Select the first K eigenvectors with the highest eigenvalues.
11. Form the principal component matrix using the selected eigenvectors.
12. Transform the standardized dataset:

$$\text{Reduced Data} = \text{Standardized Data} \times \text{Selected Eigenvectors}$$
13. Calculate the explained variance of the selected principal components.
14. Validate that the reduced dataset contains exactly K dimensions.
15. Check that the selected components preserve the major information in the original dataset.
16. Display the reduced dataset and explained variance.

END

Output

The original high-dimensional dataset is converted into a smaller dataset containing the most significant principal components.

15. Write a pseudocode algorithm to perform data validation and integrity checks on a dataset before analysis. The algorithm should verify data types, detect missing or invalid entries, enforce constraints (such as value ranges), and generate a validation report. Include steps for filtering errors, logging issues, and ensuring the dataset is ready for further processing.

Input

A dataset D that needs to be validated before analysis.

Output

A validated dataset and a validation report.

Pseudocode

BEGIN

1. Read the dataset D from the input file.
2. Check whether the file exists and can be opened.

IF file cannot be opened THEN

Log "File Input Error"

STOP

END IF

3. Check whether the dataset is empty.

IF dataset is empty THEN

Log "Dataset is Empty"

STOP

END IF

4. Check the data type of every column.

5. Identify missing or null values.

6. Identify duplicate records.

7. Check numerical columns against their permitted value ranges.

8. Check categorical columns against their allowed categories.

9. FOR each record in the dataset DO

IF data type is incorrect THEN

Log "Invalid Data Type"

END IF

IF required value is missing THEN

Log "Missing Value"

END IF

IF value is outside the permitted range THEN

Log "Invalid Range"

END IF

IF categorical value is not allowed THEN

Log "Invalid Category"

END IF

END FOR

10. Filter or remove records containing critical errors.

11. Count the following:

Total Records

Valid Records

Invalid Records

Missing Values

Duplicate Records

12. Generate a validation report containing all detected errors and their counts.

13. Check the final dataset for consistency.

14. IF no critical errors remain THEN

 Mark dataset as "Ready for Analysis"

ELSE

 Mark dataset as "Requires Correction"

END IF

15. Save the validated dataset.

16. Save the validation report.

17. Display the validation results.

END

Output

A validated dataset with a complete report of missing, duplicate, invalid, and inconsistent records.